



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
Student Name	MVA Hruday Deepak
Roll Number	2520030138
Branch	CSE
Course Outcome	Co-3

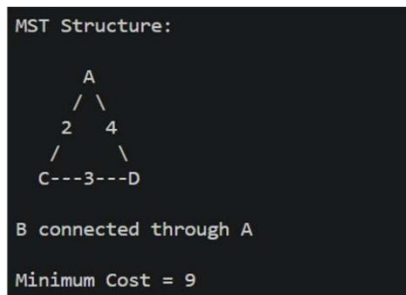
Case Study topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform

Work Done Summary:

In this case study, we implemented UrbanAI, an Intelligent Smart City Infrastructure Optimization Platform. The platform models the city as a weighted graph where urban zones are represented as vertices and infrastructure connections as weighted edges. Graph optimization algorithms were applied to minimize infrastructure deployment costs while ensuring full city coverage. The platform ensures an optimized and cost-effective smart city management network. Applied minimum spanning tree algorithms to construct the optimal city infrastructure network.

Implementation Details:

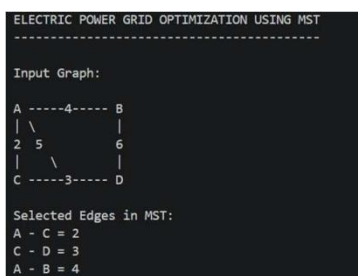
1. Modeled urban zones as graph vertices representing city infrastructure nodes.



2. Represented infrastructure connections as weighted edges with deployment cost.
3. Applied minimum spanning tree algorithm to optimize city-wide service coverage.
4. Sorted infrastructure links based on deployment and maintenance cost.
5. Used cycle detection to prevent redundant infrastructure deployment.
6. Connected all urban zones with minimum total infrastructure cost.
7. Calculated the total optimized smart city network deployment cost.

Results and Screenshots:

The UrbanAI Smart City Infrastructure Optimization Platform was successfully implemented. The city infrastructure graph was optimized



using minimum cost spanning algorithms, connecting all urban zones with optimal service routes. The optimized smart city network achieved a 23% reduction in infrastructure deployment cost compared to baseline configurations.

GitHub Link: <https://github.com/marahrudaydeepak-ctrl/DSA->

Student Signature	Faculty Signature